

David Esteban Puentes Barrero

Full-Stack Engineer | React | TypeScript | Python | AWS | Docker

Medellín, Antioquia – Colombia

☎ +57 319 392 9347 • ✉ barrerodavid10@gmail.com

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Professional Profile

Systems and Computer Engineering professional with over 3 years of experience as a Full-Stack Developer, specialized in building scalable web applications, robust APIs, and data-driven solutions.

Experienced across the full software lifecycle: requirements analysis, architecture design, frontend and backend development, cloud service integration, and deployment. Focused on delivering maintainable, efficient products aligned with real business needs.

Professional Experience

Space-Eyes

Full-Stack Developer

Medellín, Colombia

Aug 2025 – Present

Company focused on real-time monitoring, detection, and control solutions.

- Lead the frontend development of Morpheus, a real-time monitoring platform for drone detection and management, built with React and TypeScript.
- Define the frontend architecture, coding standards, and component structure, ensuring scalability, maintainability, and consistency across the development team.
- Build operational dashboards with interactive geospatial visualization using Mapbox GL JS.
- Implement reactive connectivity monitoring systems, enhancing user experience through visual feedback and dynamic interface control.
- Design and implement APIs with FastAPI for real-time telemetry processing and streaming.
- Develop Python pipelines for RF signal processing using specialized hardware such as BladeRF and HackRF One.
- Deploy serverless solutions on AWS Lambda and S3, using Docker to maintain consistency across environments.
- Optimize critical processes through selective migrations to Rust.

Projects: Morpheus, FireWatch, SeaWatch

Technologies: React, TypeScript, Python, FastAPI, Mapbox GL JS, Docker, AWS, PostgreSQL, Rust, Git, Jira

Dirección TIC – UPTC

Full-Stack Developer

Tunja, Colombia

Jul 2024 – Present

Area responsible for developing institutional technological solutions.

- Led end-to-end development of a predictive platform for academic space management and allocation, optimizing resources and automating administrative processes.
- Designed and implemented the frontend using Angular and TypeScript, building complex modules with a focus on usability and visual clarity.
- Implemented a centralized notification system using ToastService and PrimeNG.
- Developed a global HTTP error interceptor to handle failures consistently and improve user experience.
- Migrated the styling structure from SCSS to plain CSS, reducing complexity and improving maintainability.
- Standardized backend communication through centralized endpoint configuration.
- Participated in backend service integration with Spring Boot and authentication with Keycloak.

Technologies: Angular, TypeScript, PrimeNG, Spring Boot, Docker, GitHub Actions, Keycloak

- Developed interactive dashboards for real-time business metrics monitoring, supporting data-driven decision making.
- Designed and implemented REST APIs in Python for service integration and information processing.
- Automated client report generation processes, reducing repetitive manual tasks.
- Participated in the development of web applications focused on management, security, and information analysis.
- Worked with SQL databases for data modeling, querying, and administration.

Technologies: Python, REST APIs, SQL, Angular, C#, Git

Education

Technical Skills

Frontend: React, Angular, Next.js, Astro, JavaScript, TypeScript, Vite, Tailwind CSS, Mapbox GL JS
Backend: Python, FastAPI, Django, Node.js, NestJS, Java, Spring Boot, Rust
Databases: PostgreSQL, MySQL, SQL Server, Oracle
Cloud & DevOps: AWS, Docker, CI/CD, GitHub Actions
Tools: Git, Jira, Linux, Postman, Figma

Certifications

Anthropic <i>Claude Code in Action</i>	2026
Platzi <i>Frontend Developer</i>	2026
Platzi <i>AWS Fundamentals</i>	2026
Platzi <i>Machine Learning</i>	2025

Languages

Spanish: Native
English: B2 – Upper Intermediate